

Smart Grocery Inventory, AI Recipes & Meal Planner System

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INTRODUCTION

Modern households struggle with tracking groceries, planning meals, and managing shopping lists efficiently.

Manual methods lead to

- Forgotten items
- Unplanned meals
- Food wastage
- Inefficient shopping

GrocyGo is a **multi-platform system** (Web + Mobile + Backend + AI) designed to **automate** grocery tracking, meal suggestions, and shopping list creation.

OBJECTIVES

- Track groceries with **quantity**, **expiry**, and **location**
- Suggest **AI-powered** recipes based on **8000+** dataset
- Auto-generate smart shopping lists
- Plan weekly meals (breakfast/lunch/dinner)
- Reduce **food waste** with **expiry alerts**
- Provide analytics for usage and shopping patterns
- Build a scalable multi-platform system (**React + Flutter + Node.js**)

RESULTS & DISCUSSION

Key Outcomes

- Automated tracking reduced manual effort
- AI suggestions improved meal decision-making
- Shopping list accuracy increased with auto-generated items
- Expiring items identified early → reduced waste
- Analytics provided insights into category usage & purchase behavior

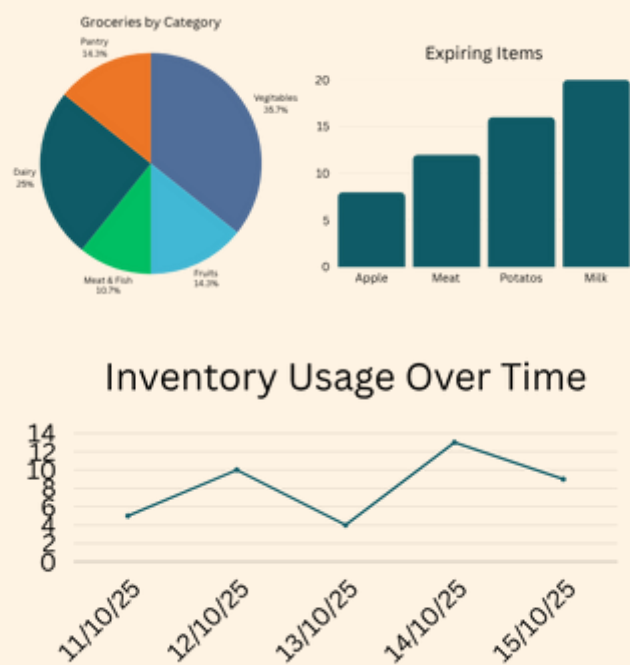
Visual Results

Include:

- Pie chart (Grocery categories)
- Bar chart (Expiring items per month)
- Line graph (Usage trends)
- Table or screenshot examples
- Small analytics dashboard preview

Performance

- AI suggestions responded < 1 sec
- Smooth UI across web and mobile
- Efficient dataset filtering



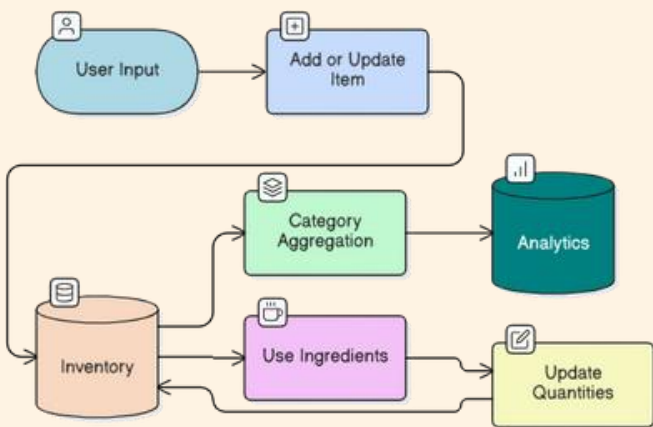
METHODOLOGY

Tools & Technologies

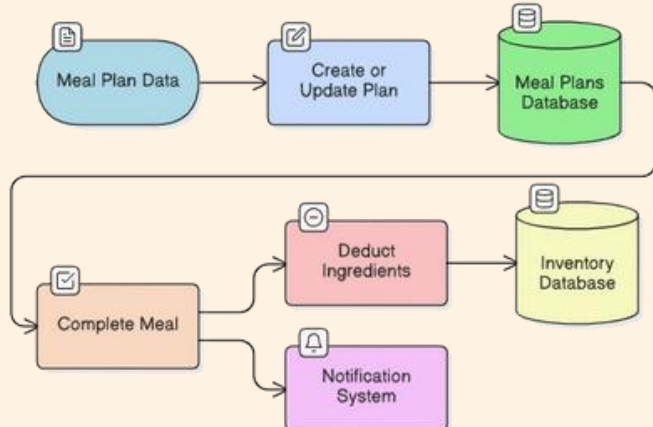
- Frontend: React.js (Web), Flutter (Mobile)
- Backend: Node.js + Express
- AI Engine: Dataset-based recipe filtering
- Dataset: 8000+ recipes (CSV)
- Charts: React charting libraries
- Architecture: MVC + REST API

Process Flow

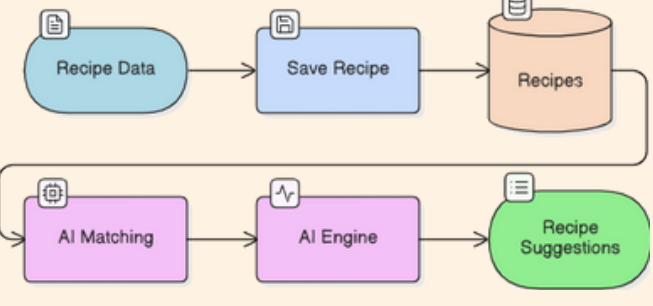
• Inventory Management



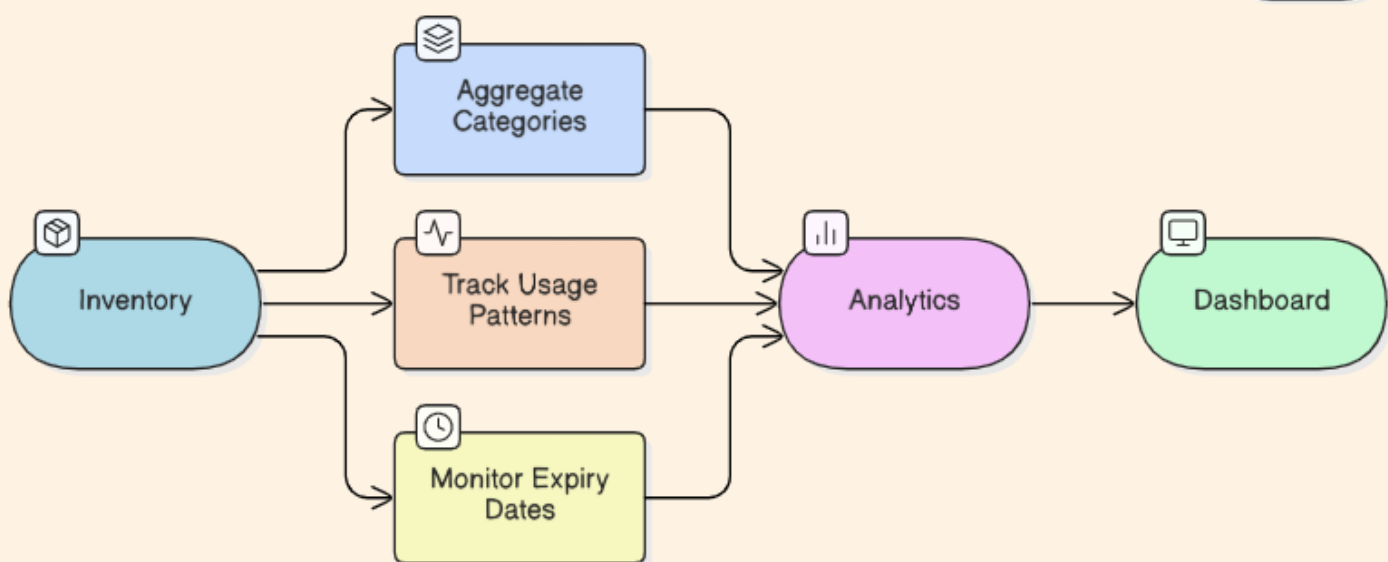
• Meal Planning



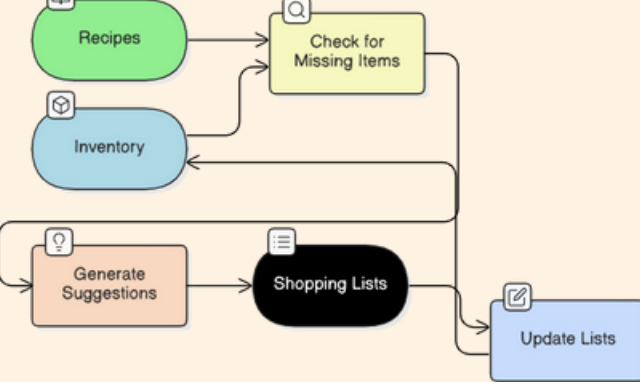
• Recipe Management



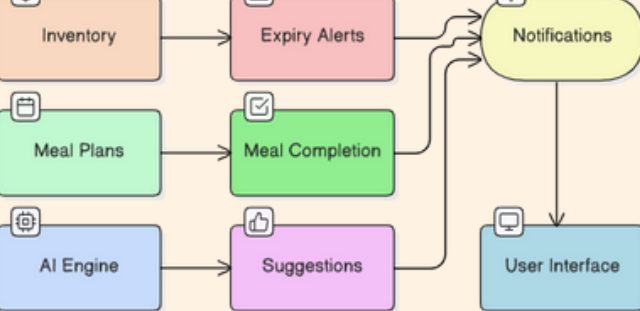
• Analytics Generation



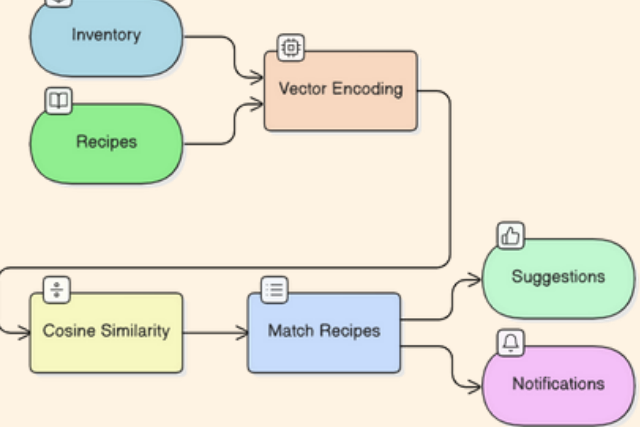
• Shopping List Management



• Notification Management



• AI Suggetion Engine



CONCLUSION

GrocyGo successfully provides an intelligent, user-friendly solution for household grocery management.

It integrates Web, Mobile, Backend, and AI to deliver:

- Smart inventory
- Efficient meal planning
- Accurate shopping lists
- Useful analytics
- Reduced food wastage

Future Enhancements

- Barcode/QR scanning
- Voice assistant integration
- Cloud database & real-time sync
- AI nutritional analysis
- Auto grocery ordering

Applications

- Smart kitchens
- Hostel/PG food management
- Small restaurants
- Family grocery coordination

Limitations

- Dataset-based AI (not deep-learning)
- No cloud-based multi-user sync yet